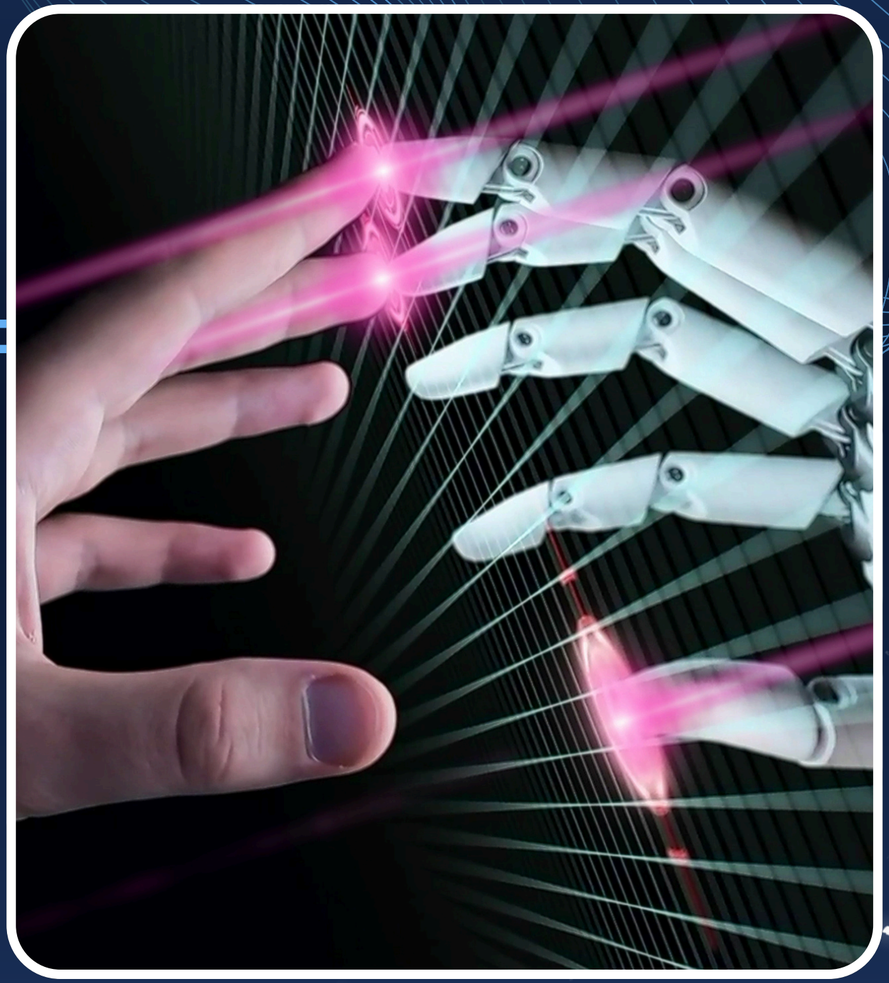
VAYU SURYA AI



An AI Solution by Team VSA

PROBLEM STATEMENT

AI in Renewable Forecasting – Key Challenges

Weather variability → inaccurate predictions & grid imbalance

Legacy systems → limited modification possible

Need: accurate, explainable forecasts (day-ahead to hourly)

Data issues → masked/synthetic data, LLM restrictions

Failures due to poor data, wrong models, weak infra & integration

Risks: cybersecurity, low storage, skill gaps

Impact: low accuracy, financial losses, unreliable systems

SOLUTION

AI forecasting assistant for solar & wind plants

Learns plant behavior under changing weather

Provides hourly, intra-day & day-ahead predictions

Gives confidence ranges (uncertainty included)

Explainable insights (e.g., clouds, low wind reasons)

No black-box → transparent & trustworthy

No system overhaul required → easy integration

ABOUT PROJECT

: Project Overview

AI-based forecasting system for solar & wind power plants

Uses weather forecasts + historical generation data

Generates hourly, intra-day & day-ahead predictions

Provides probabilistic outputs (p10/p50/p90) with uncertainty ranges

Explainable insights using key drivers (cloud cover, wind speed, etc.)

No black-box → transparent, reliable, and planner-friendly

Designed to work without modifying existing infrastructure

: System Design & Impact

Secure deployment: runs on-prem/local (no external data sharing)

Synthetic data used for development → ensures privacy compliance

Data pipeline: Input (weather + generation) → processing → AI model → calibrated output

Handles missing data & ensures robust performance

Outputs in JSON/CSV + simple human-readable explanations

24/7 low-latency availability via API/dashboard

Impact: improved scheduling, reduced curtailment & losses, better grid stability

HOW IT WORK

: How It Works (Simple View)

Works like a smart plant operator that never sleeps

Learns patterns for each solar plant & wind cluster

Monitors weather (clouds, wind) + past power generation

Predicts future power output (hourly, intra-day, day-ahead)

Gives a range (e.g., likely between X–Y) instead of a single value

Explains reasons behind changes (cloud cover, low wind, etc.)

: How It Works (Technical Flow)

Input: Weather forecasts + historical generation data (CSV/API)

Processing: Data cleaning + feature engineering

Model: Probabilistic AI (quantile regression + conformal prediction)

Output: p10/p50/p90 forecasts with uncertainty bands

Explainability: Key drivers identified using SHAP

Delivery: Results via API/dashboard in JSON/CSV format

FAILURE FACTORS IN AI RENEWABLE FORECASTING

Poor data quality & inconsistent datasets

Highly unpredictable weather conditions

Incorrect model selection

Lack of real-time infrastructure

Weak grid integration

Shortage of skilled workforce

Cybersecurity risks

Lack of energy storage systems

Result: low accuracy, financial losses, unreliable performance

MAIN ISSUES OF THE THEME

Renewable energy highly weather-dependent → variability & grid imbalance

Scheduling inefficiencies due to uncertain generation

Existing systems cannot be modified

Need for accurate & explainable forecasts (hourly to day-ahead)

Uncertainty must be quantified

Data constraints: masked/synthetic data, no hosted LLMs

CONCLUSION

Our AI solution provides accurate, explainable & uncertainty-aware predictions

Works securely within existing infrastructure (no major changes needed)

Improves scheduling, reduces losses & enhances grid stability

Scalable solution for better renewable integration in the future

Renewable forecasting faces challenges due to variability, data limits & legacy systems